Remaining Work — Group 20

AMS 560 · Big Data Systems, Algorithms and Networks · Fall 2026

"Prioritize resolving integration issues and completing end-to-end comparisons." — Professor

---

Quick Status — All Members

Member	Role	Code Status	Blocking Item
Jyotir	Filter + Baselines	**Complete + bug fixed**	Baseline decision logs (needs Riya's grid)
Krupesh	Router + Integration	**Complete**	2 adapter bugs to fix before standalone run
Parikshit	Simulator + Traces	**Complete**	node.py crash on KV overflow (fix before final grid)
Sur	LinUCB + Reward + Sweep	**Complete**	Team reward-weights decision for S3
Riya	Experiments + Report	**Pipeline complete**	Run 420-grid; write Eval/Results/Discussion

The end-to-end pipeline is built and has run. The professor's remark drives a strict priority: (1) fix the two integration bugs that break standalone runs, (2) fix the simulator crash, (3) run the full 420-run experiment grid, (4) generate figures and write the final report.

Jyotir — Filter and Baselines

All deliverables complete. One regression bug was identified by Sur in RESULTS.md and fixed on 2026-11-10.

Bug fixed (2026-11-10): kv_cache_max_frac not enforced in token-exact path

The token-exact KV memory check in `filter.py` Rule 3 compared the request's token footprint against 100% of capacity instead of the configured `kv_cache_max_frac` (0.90). A node at 85% fill would accept a request that pushed usage to 95%, violating the safety headroom.

- File: `feasibility_filter/filter.py`, Rule 3 (Memory)
- Fix: `kv_used_tokens + tokens_needed > kv_capacity * kv_limit` (was: `> kv_capacity`)
- New test: `TestTokenAwareMemoryCheck::test_token_check_respects_kv_cache_max_frac`
- All 28 tests pass.

Remaining open item

Baseline decision logs — running all 6 baselines on all 6 scenarios to produce the Parquet/CSV logs Riya needs. No additional coding required: the baselines are already registered as `jyotir:*` in Krupesh's registry and are included in `config_simulator.json`. Logs will be produced as a side effect of Riya's full 420-run grid once the pipeline is green.

Coordination items

Item	Owner	Priority
Fix stale F1/F2 bug comments in jyotir_adapter.py lines 13–16 (both bugs already fixed)	Krupesh	Medium
Fix default JYOTIR_DIR resolving to wrong path in jyotir_adapter.py line 29	Krupesh	High
Optional: Sur spot-checks S2/S4 after kv_cache_max_frac fix to confirm sweep numbers stable	Sur	Low
Team decision: reward weights (energy scale) before final runs	Team	High

Krupesh — Router Service and Integration Layer

Router service, decision log, policy interface, and integration adapter are complete. Two bugs in `jyotir_adapter.py` must be fixed before the router can be run standalone (without Riya's path override).

Bug A — Stale F1/F2 comments (documentation bug)

File: `integration/jyotir_adapter.py`, lines 13–16

Comments document F1 (warming node skips safety rules) and F2 (memory ignores request size) as "known behaviour". Both bugs were fixed by Jyotir (commit 59b63af). A grader reading the adapter will think the bugs still exist.

Fix: replace the F1/F2 comment block with: `"Jyotir's filter as of commit 59b63af+: warming nodes go through all 5 rules; memory rule enforces kv_cache_max_frac."`

Bug B — Default JYOTIR_DIR resolves to wrong path (runtime bug)

File: `integration/jyotir_adapter.py`, line 29

```
JYOTIR_DIR = Path(__file__).resolve().parents[2] / "Jyotir"
```

Because Krupesh's folder is uploaded one level deeper than the repo root, `parents[2]` resolves to `Krupesh/geo-inference-scheduler/` → `JYOTIR_DIR` becomes `Krupesh/geo-inference-scheduler/Jyotir/` which does not exist. Running `uvicorn router_service.app:app` standalone fails at import time. Works only because Riya's runner overrides this variable before importing.

Fix: walk up to the repo root using a known sentinel file (e.g. the `Parikshit/` folder), or hard-code `parents[4] / "Jyotir"` to match the actual folder depth.

Optional improvement — telemetry-age accounting

Sur's sweep found ~3,200 true-state power violations per S2 run with the per-node W=500 policy, all from the router's 1-second-old telemetry. The router already counts its in-flight placements against load; applying the same accounting to power draw and temperature would reduce this. Requires Parikshit's per-request power delta. Team decision before final runs.

Parikshit — Simulator and Event Traces

Simulator, telemetry schema, all 6 scenario traces, 122 tests, and the KrupeshRouter integration bridge are complete. One crash bug must be fixed before the final grid runs.

Bug — node.py crashes on KV overflow instead of calling on_outcome

File: `simulator/node.py`, lines 178–179

```
if job.kv_tokens > self.kv_capacity:
raise ValueError(...)
```

If a request slips past the filter with more tokens than the node's total KV capacity (possible with stale telemetry during a burst), the simulator raises an uncaught `ValueError` and crashes the whole run, losing all results.

Fix: call the failure callback instead of raising — the same pattern as the `go_down()` path:

```
if job.kv_tokens > self.kv_capacity:
self.on_outcome(self._failure(job, self.env.now))
return
```

Priority: **High** — this would crash Riya's 420-run grid if any run hits the edge case.

Optional / team decisions

Item	Who decides	Impact if changed
S2 burst hardness: raise base rate so 3× burst exceeds cluster max throughput	Team	Sweep no longer directly comparable; re-run needed
Telemetry tick: reduce 1 s → 0.5 s to halve true-state staleness	Team + Krupesh	Minor performance change; must coordinate with router

Sur — LinUCB Policy, Reward, and Window Sweep

All code, the 1,350-run window sweep on the real pipeline, and the full set of sweep tables and figures are complete. One team decision blocks finalising the S3 results.

Team decision required: energy weight for S3 price-aware routing

With current reward weights (SLO = 1.0, energy = 0.1, `energy_cost_max_usd = $0.00001`), the learned policy gives **identical results in S3 (price shift) and S1 (stable)**. The energy term is too small relative to SLO: a typical request's energy contribution moves the reward by only ~0.01, so the policy never shifts traffic to cheaper sites when California's prices jump to $0.046–0.050/kWh.

Option	Change	Effect on S3	Effect on S1 goodput
Keep current weights	None	No price-aware routing; lowest-energy is only realistic baseline	72.2 req/s (unchanged)
Raise energy weight	Increase energy YAML weight or raise `energy_cost_max_usd`	Policy shifts to cheaper nodes after price jump	May lose ~1–2 req/s (SLO no longer fully dominates)

Config file: `Sur/configs/policies/sw_linucb.yaml`. Decision must be made **before Riya's final grid run**. Changing after the run means re-running.

Optional: spot-check after Jyotir's kv_cache_max_frac fix

The window sweep ran with Jyotir's pre-fix filter. The fix tightens the memory check by 10% of capacity. Sur's RESULTS.md notes no S6 true-state violations involved the warming node and the fix only affects nodes between 90–100% KV fill. A quick 1-seed spot check on S2 and S4 (most true-state violations) is recommended before the final grid to confirm sweep numbers are stable.

Sweep findings for the report (key numbers)

Finding	Number
Learned policy (W=500 per-node) vs best baseline (lowest-energy) in S1 goodput	72.2 vs 61.1 req/s
Adaptation time S2: W=500 per-node vs plain LinUCB	20.4 s vs 33.6 s [95% CI]
Hard-rule violations by filter-allowed nodes (all 1,350 runs)	0
True-state power violations per S2 run (W=500)	~3,200 (telemetry age, not policy fault)
Cost per SLO-met request S3: W=500 vs round-robin	268 µ$ vs 677 µ$

Riya — Experiments, Metrics, and Final Report

The full end-to-end experiment pipeline is complete: simulator → router → filter → policy → decision log → metrics → recovery → bootstrap CIs → figures. The 420-run grid has not yet been executed. Report Evaluation, Results, and Discussion sections are not yet written.

Step 1 — Pipeline validation (run first)

```
python run_evaluation.py --config config_validation.json --output ../results/validation_01
```

Runs 1 policy × 3 scenarios × 1 seed (3 runs, ~5–10 min). Confirms `true_state_violation` column present, all bundles status=ok, and `decision_log.csv` passes `validate_log()`. **Requires Parikshit's node.py crash fix first.**

Step 2 — Full 420-run grid

```
python run_evaluation.py --config config_simulator.json --output ../results/final_01
```

7 policies × 6 scenarios × 10 seeds = 420 runs. Expected ~21 min on 16 processes. Use `--resume` if interrupted. After run: check `run_coverage.json` shows complete=true and `pairing_checks.csv` shows workloads_match=true for all seeds. **Requires team reward-weights decision first.**

Step 3 — Figures

```
python figures/plot_results.py --results ../results/final_01
```

Produces goodput-over-time curves and 95% bootstrap CI bar charts for 9 key metrics. Requires ≥10 seeds (all 10 needed for `ci_status=available`).

Step 4 — Report sections to write

Section	Inputs needed	Status
Evaluation setup	config_simulator.json (already frozen)	Not started
Results — goodput & adaptation	final_01/figures/ and recovery_summary.csv	Not started
Results — cost and energy	confidence_intervals.csv	Not started
Ablation runs (3 configs)	no-filter, no-context, no-forgetting variants	Not started
Discussion	After results sections complete	Not started
Final report assembly	All 5 member sections + above	Not started

Three reporting caveats to include: (1) Lost goodput is null until a matched oracle (round-robin baseline at same seed) is passed to recovery metrics. (2) Report every adaptation time against the S1 stable-noise floor (Sur found 4–12 s pseudo-delays in 4/10 seeds). (3) S6 needs two anchor times: t0=200 s (outage) and t0=400 s (node return), with the returned node's traffic share shown alongside the adaptation time.

Critical Path to Final Results

These items must be done in order. All other remaining items are parallel or optional.

#	Action	Owner	Blocks
1	Fix node.py crash (raise → on_outcome callback)	Parikshit	Riya's grid
2	Fix jyotir_adapter.py JYOTIR_DIR default path	Krupesh	Standalone router; Riya's pipeline
3	Team reward-weights decision for S3 (energy scale)	Team	Meaningful S3 results
4	Run config_validation.json (3-run smoke test)	Riya	Full grid confidence
5	Run full 420-run grid with config_simulator.json	Riya	Figures, results, paper
6	Generate figures (plot_results.py)	Riya	Paper figures
7	Write Evaluation + Results + Discussion + Conclusion	Riya	Final submission
8	Assemble final report (all 5 sections)	Riya	Submission

Parallel (do not block the critical path):

Action	Owner	Priority
Fix stale F1/F2 comments in jyotir_adapter.py lines 13–16	Krupesh	Medium
Optional: Sur spot-check S2/S4 after Jyotir's kv_cache_max_frac fix	Sur	Low
Optional: team discussion on S2 burst hardness and telemetry tick interval	Team	Low
Optional: provide round-robin oracle curve for lost-goodput metric	Riya	Low

---

Generated 2026-11-10. All individual code deliverables are complete. Critical path: Parikshit's crash fix → reward-weights decision → Riya's 420-run grid → figures → paper.